Geometric Quantities for the Schwarzschild Metric

Generated by Einsteinpy_to_latex

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1 Citation Information

If you use these results and code in your research, please cite:

- Einstein, A. (1915). Die Feldgleichungen der Gravitation. *Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften*, 844-847.
- Ribeiro, S., Bapat, A., Tandon, A., & EinsteinPy Developers (2020). EinsteinPy: A Python Package for General Relativity. *Journal of Open Source Software*, 5(51), 2356. `https://doi.org/10.21105/joss.02356`
- P. Ntelis (2026). Einsteinpy_to_latex: Schwarzschild Metric Tensor Calculator (Version 1.0). `https://github.com/lontelis/Einsteinpy_to_latex`

2 Geometric quantities for the Schwarzschild metric

The Schwarzschild metric is given by

$$ds^2 = -\left(1 - \frac{2GM}{c^2 r}\right) c^2 dt^2 + \left(1 - \frac{2GM}{c^2 r}\right)^{-1} dr^2 + r^2 d\theta^2 + r^2 \sin^2\theta d\phi^2. \quad (1)$$

2.1 Metric tensor $g_{\mu\nu}$

$$g_{\mu\nu} = \begin{bmatrix} -c^2\left(-\frac{2GM}{c^2 r} + 1\right) & 0 & 0 & 0 \\ 0 & \frac{1}{-\frac{2GM}{c^2 r} + 1} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2(\theta) \end{bmatrix} \quad (2)$$

2.2 Christoffel symbols $\Gamma^{\mu}_{\nu\rho}$ (non-zero)

$$\Gamma^{t}_{tr} = \frac{GM}{r\left(-2GM + c^{2}r\right)} \tag{3}$$

$$\Gamma^{t}_{rt} = \frac{GM}{r\left(-2GM + c^{2}r\right)} \tag{4}$$

$$\Gamma^{r}_{tt} = \frac{GM\left(-2GM + c^{2}r\right)}{c^{2}r^{3}} \tag{5}$$

$$\Gamma^{r}_{rr} = \frac{GM}{r\left(2GM - c^{2}r\right)} \tag{6}$$

$$\Gamma^{r}_{\theta\theta} = \frac{2GM}{c^{2}} - r \tag{7}$$

$$\Gamma^{r}_{\phi\phi} = \frac{\left(2GM - c^{2}r\right)\sin^{2}(\theta)}{c^{2}} \tag{8}$$

$$\Gamma^{\theta}_{r\theta} = \frac{1}{r} \tag{9}$$

$$\Gamma^{\theta}_{\theta r} = \frac{1}{r} \tag{10}$$

$$\Gamma^{\theta}_{\phi\phi} = -\frac{\sin(2\theta)}{2} \tag{11}$$

$$\Gamma^{\phi}_{r\phi} = \frac{1}{r} \tag{12}$$

$$\Gamma^{\phi}_{\theta\phi} = \frac{1}{\tan(\theta)} \tag{13}$$

$$\Gamma^{\phi}_{\phi r} = \frac{1}{r} \tag{14}$$

$$\Gamma^{\phi}_{\phi\theta} = \frac{1}{\tan(\theta)} \tag{15}$$

$$\tag{16}$$

2.3 Riemann curvature tensor $R^{\mu}{}_{\nu\rho\sigma}$ (non-zero)

$$R^{t}_{rtr} = \frac{2GM}{r^2\left(-2GM + c^2 r\right)} \tag{17}$$

$$R^{t}_{rrt} = \frac{2GM}{r^2\left(2GM - c^2 r\right)} \tag{18}$$

$$R^{t}_{\theta t\theta} = -\frac{GM}{c^2 r} \tag{19}$$

$$R^{t}_{\theta\theta t} = \frac{GM}{c^2 r} \tag{20}$$

$$R^{t}_{\phi t\phi} = -\frac{GM\sin^2\left(\theta\right)}{c^2 r} \tag{21}$$

$$R^{t}_{\phi\phi t} = \frac{GM\sin^2\left(\theta\right)}{c^2 r} \tag{22}$$

$$R^{r}_{ttr} = \frac{2GM\left(-2GM + c^2 r\right)}{c^2 r^4} \tag{23}$$

$$R^{r}_{trt} = \frac{2GM\left(2GM - c^2 r\right)}{c^2 r^4} \tag{24}$$

$$R^{r}_{\theta r\theta} = -\frac{GM}{c^2 r} \tag{25}$$

$$R^{r}_{\theta\theta r} = \frac{GM}{c^2 r} \tag{26}$$

$$R^{r}_{\phi r\phi} = -\frac{GM\sin^2\left(\theta\right)}{c^2 r} \tag{27}$$

$$R^{r}_{\phi\phi r} = \frac{GM\sin^2\left(\theta\right)}{c^2 r} \tag{28}$$

$$R^{\theta}_{tt\theta} = \frac{GM\left(2GM - c^2 r\right)}{c^2 r^4} \tag{29}$$

$$R^{\theta}_{t\theta t} = \frac{GM\left(-2GM + c^2 r\right)}{c^2 r^4} \tag{30}$$

$$R^{\theta}_{rr\theta} = \frac{GM}{r^2\left(-2GM + c^2 r\right)} \tag{31}$$

$$R^{\theta}_{r\theta r} = \frac{GM}{r^2\left(2GM - c^2 r\right)} \tag{32}$$

$$R^{\theta}_{\phi\theta\phi} = \frac{2GM\sin^2\left(\theta\right)}{c^2 r} \tag{33}$$

$$R^{\theta}_{\phi\phi\theta} = -\frac{2GM\sin^2\left(\theta\right)}{c^2 r} \tag{34}$$

$$R^{\phi}_{tt\phi} = \frac{GM\left(2GM - c^2 r\right)}{c^2 r^4} \tag{35}$$

$$R^{\phi}_{t\phi t} = \frac{GM\left(-2GM + c^2 r\right)}{c^2 r^4} \tag{36}$$

$$R^{\phi}_{rr\phi} = \frac{GM}{r^2\left(-2GM + c^2 r\right)} \tag{37}$$

$$R^{\phi}_{r\phi r} = \frac{GM}{r^2\left(2GM - c^2 r\right)} \tag{38}$$

$$R^{\phi}_{\theta\theta\phi} = -\frac{2GM}{c^2 r} \tag{39}$$

$$R^{\phi}_{\theta\phi\theta} = \frac{2GM}{c^2 r} \tag{40}$$

$$\tag{41}$$

2.4 Ricci tensor $R_{\mu\nu}$ (non-zero)

$$Allcomponentszero. \tag{42}$$

2.5 Ricci scalar R

$$0 \tag{43}$$

2.6 Einstein tensor $G_{\mu\nu}$ (non-zero)

$$Allcomponentszero. \tag{44}$$